

What Does it Mean for Water to Return?: A Degrowth Proposal Addressing Flooding in Germantown, PA



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I. Introduction

Hidden under Philadelphia's concrete surface is a historic secret: 283 linear miles of streams.¹ As Philadelphia grew into the large metropolis it is today, these waterways were built over, culverted into pipes used to dispose of household and industrial waste. With heavy rain events, these waterways naturally expand as they would above ground, overwhelming the outdated infrastructure that encases them and causing serious and frequent flooding events in neighborhoods such as Germantown. What might it mean for these waterways to resurface? Building from conversations through the WaterShed, a community hub in Germantown offering resources and education about local environmental challenges, particularly with community leader Reverend Chester Williams, we have developed a just-transition oriented proposal for addressing flooding in Germantown at its root. We use degrowth as the lens through which we propose our vision of a just-transition, understanding degrowth broadly as a prioritization of quality of life over economic growth. This proposal is supported by historic and theoretical context, in which we explore racial capitalism, gentrification in Germantown, zoning issues and an alternative to zoning. We propose a two-pronged approach: a mutual aid program aimed at addressing community member's immediate needs, and the daylighting of streams that run under vacant lots throughout Germantown and the broader Philadelphia region. In the context of the government's long-standing repression of radical alternatives, this proposal is primarily aimed toward Germantown residents and anyone interested in radical alternatives to the status quo systems prioritizing economic growth above all, which are only concealing and exacerbating the dire threats of flooding.

A. *History of Germantown*

The Philadelphia Water Department describes the history of Germantown and flooding as beginning with Philadelphia's initial urban boom in the 19th century. Throughout the 19th and 20th century, as Philadelphia grew into a large metropolitan city, the numerous waterways that flowed across the city were used to dispose of household and industrial waste. These streams and creeks became heavily polluted, transforming into "open-air sewers," and posing large scale

¹ Justin Martin, "Philadelphia's Watershed History," August 4, 2023, <https://water.phila.gov/watersheds/history/>.

public health risks for the surrounding community through the spread of diseases like cholera and typhoid fever.²

In the attempt to mitigate the health hazard of raised sewer systems, the city of Philadelphia converted many of the polluted streams into closed sewer pipes. One of these covered streams, Wingohocking Creek, historically ran from Mount Airy, across Germantown, and emptied into the Frankford Creek located in current day Juniata Park. The City enclosed and paved over the majority of this creek in a project that lasted from 1879 to the 1940s, and turned the stream into a giant sewer and stormwater hold. This infrastructure was only updated following guidelines of the Clean Water Act of 1972 that resulted in new water treatment processes but maintained the same pipe systems. The outdated infrastructure running through the city directly impacts residents and plays a role in current day flooding and overflow sewer runoff to the surrounding waterways.³



Fig. 1 Outfall of Wingohocking Sewer, dated December 20, 1954. (From the City Archives of Philadelphia)

As the neighborhoods of both Mount Airy and Germantown continued to grow, the system became overwhelmed; now, more than a century later, the sewer line is unable to support the population size. Although the flooding in Northwest Philadelphia is particularly severe, the issue of strained sewer lines is a constant throughout the city. The vast network of covered

² Amy Hopf, "Germantown Flooding: Community Connections to the Past, Present, Future," Philadelphia Water Department, February 24, 2025. <https://water.phila.gov/blog/germantown-flooding-task-force/>.

³ Adam Levine, "Sewers, Pollution, and Public Health in 19th Century Philadelphia," January 22, 2025, <https://waterhistoryphl.org/articles/sewers-pollution-public-health/>.

streams now dumps billions of gallons of stormwater mixed with raw sewage into the local waterways each year.⁴

1. Flood insurance

The preventative and management options for flooding concerns in Germantown are not only insufficient but are often inaccessible. Beyond infrastructural updates, it is recommended that people invest in flood insurance, something rarely included in standard homeowners insurance plans. Philadelphia residents are all eligible to apply for FEMA's National Flood Insurance Program (NFIP).⁵ With a NFIP insurance plan, homeowners, renters, and business owners can receive support in recovering from flood damage. However, options such as these are currently inadequate and inaccessible. Firstly, when people are already struggling financially, the cost of adding flood insurance is unattainable. However, the inaccessibility of flood insurance is further compounded by the issue of tangled titles. "Tangled Titles" explain the circumstance of a person living in a home where they do not hold the deed to the property in their name. These situations often arise from multi-generational families living in the same home without going through the legal process of converting ownership. At least 2% of residential properties in Philadelphia are considered tangled titles. Being in such a situation can cause anxiety around receiving aid from the government, and other processes necessitates their properties' legal status to be inspected. The implications for being discovered could lead to eviction. It is critical to note that those with tangled titles tend to be "those with relatively low housing values, low incomes, and high poverty rates... [with] predominantly Black [populations]."⁶ Tangled titles prevent people from receiving potential resources in aiding flood prevention measures; those who most need it might be struggling with a series of issues, including tangled titles. Therefore, there are a series of nuances that must be considered to understand the difficulties presented by flooding.

⁴ Penn Environment Research and Policy Center, "Sewage Pollution in the Philadelphia-Camden Region," October 27, 2025,

<https://environmentamerica.org/pennsylvania/center/resources/sewage-pollution-in-the-philadelphia-camden-region>.

⁵ "Philadelphia's Acceptance into FEMA Community Rating System Opens Doors for Discounts on Flood Insurance Policies: Flood Management," City of Philadelphia, January 7, 2026,

<https://www.phila.gov/2026-01-07-philadelphias-acceptance-into-fema-community-rating-system-opens-doors-for-discounts-on-flood-insurance-policies/>.

⁶ "How 'Tangled Titles' Affect Philadelphia," How "Tangled Titles" Affect Philadelphia | The Pew Charitable Trusts, August 17, 2021, <https://pew.org/3i6McmP>.

2. *History of Degrowth in Philadelphia*

Parallels between the urban activism of the 70s and 80s with contemporary degrowth principles are exemplified in the MOVE Organization, founded in 1972 by John Africa in Philadelphia. Core to MOVE was the “sanctity of life and dismantling institutions of state and capital (including economic growth).”⁷ Fiscella argues that some of the many contributions of John Africa and MOVE to the degrowth movements were in teaching and practicing a form of degrowth in the 70s, situating “common ground between indigenous peoples (e.g. Buen Vivir, Ubuntu, AIM) and radical Black ecology together with degrowth principles through an ecocentric ontology,” “ascribing sacred status to a non-dualist conception of nature,” utilizing a decolonial semiotics to emancipate terms, connected degrowth-aligned principles to relatively non-prioritized issues, “engaged in sustained, communal resistance to state power through resilient organization,” and stitched together a broad variety of degrowth-oriented projects.⁸ In 1985, MOVE’s home and headquarters was surrounded by police after complaints from MOVE’s neighbors, resulting in police firing more than 10,000 rounds of ammunition into the building and dropping a bomb on the building. This resulted in an entire city block being burned down, which left more than 250 people homeless and killed 11 MOVE members, including five children.⁹

The history of MOVE in Philadelphia is important to contextualize this proposal and exploration of degrowth in Germantown, PA. Both the context of the longstanding history of degrowth-oriented activism in urban environments as well as the history of the city’s violent police attack on MOVE as a collective based in radical Black ecology situates degrowth as opposed to the status quo desires of the city. Systems of capitalism and growth are intrinsic to traditional government organization and legibility in American cities, and opposition to these systems is not often viewed or treated kindly by those in power. In this context, this proposal is primarily aimed toward the residents of Germantown and others interested in radical alternatives to the status quo systems of growth, development, and gentrification that are only covering up and exacerbating flood threats.

⁷ Anthony T Fiscella, “Forgotten pioneers in degrowth: John Africa and the MOVE Organization,” *Journal of Political Ecology* 29, no. 1 (June 19, 2022): 405–29, <https://journals.librarypublishing.arizona.edu/jpe/article/id/5123/>.

⁸ *Ibid.*, 421.

⁹ *Ibid.*, 415.

B. *Definitions*

1. Capitalism

Capitalism is an ideological framework that infiltrates every aspect of our lives within the United States. It has been made possible through a history and continuation of colonialism, racist institutions, and the exploitation of environmental resources both within America and on a global scale. Our proposal is grounded in the belief that capitalism can not be disentangled from racist, colonialist, and human-exceptionalist values. We will briefly explain why capitalism is at odds with environmental sustainability. Then, we will move towards an exploration into the concept of racial capitalism to contextualize economic growth within Germantown.

2. Degrowth

At its core, degrowth proposes a culture shift from the current fetishization of economic growth. This obsession manifests in everything from overconsumption and ever-present advertising, house flipping and fast and shoddy developments, to the desperate search for technological solutions that will allow for continued growth. Degrowth academic Professor Jason Hickel defines degrowth as “a planned reduction of aggregate resource and energy use in high-income nations designed to bring the economy back into balance with the living world in a safe, just and equitable way.”¹⁰ On a global scale, degrowth specifies that post-growth policies and ideologies must be adopted in the wealthy, core countries of the Global North that objectively do not need further growth and should instead “reorganize the economy around provisioning for human needs and well-being, reducing inequality and guaranteeing universal access to public goods and services...[and] scale down unnecessary forms of production and consumption.”¹¹ Degrowth is also fundamentally tied to decolonization, confronting the long and continued history of resource, labor, and land exploitation in the Global South to feed economic growth in the Global North. However, degrowth is not just a lofty, global-scale, radical proposal; much of degrowth theory is directly rooted in a call for localization and the adoption of degrowth

¹⁰ Jason Hickel and Stéphane Hallegatte, “Can We Live Within Environmental Limits and Still Reduce Poverty? Degrowth or Decoupling?” *Development Policy Review* 40, no. 1 (September 7, 2021), <https://doi.org/10.1111/dpr.12584>.

¹¹ *Ibid.*, 4.

on a local-scale to support community resilience in the face of climate change.¹² While localization and the image of a self-sufficient community might call to mind a tightknit rural town, cities have been core to the degrowth imaginary before degrowth even came into its own.

3. Mutual aid

Mutual aid is the process of creating networks of support and community care in the face of Western capitalist modes of competition. Whereas structures like charity move from top-down, presenting one group as the authority and the other as reliant, mutual aid takes a more horizontal approach, placing all participants on equal ground. Mutual aid operates with the understanding that the individual is a component of the larger society, and that by helping each other, we all can thrive. When the state refuses to act on its obligation to provide stability and services for its population, we, as community members, should aim to support the people over the system. In reference to the definitions of degrowth, mutual aid stands to create a reclamation of safety and care from the state. Growing out of the context of the devastating MOVE bombing, we aim to endorse practices of mutual aid amongst the residents of Germantown to combat the historical actions of the state to preserve individualized power over the collective.

4. Flooding

On a global level, flooding is a growing threat towards humanity. As the Earth continues to warm due to fossil-fueled climate change, the frequency of flooding is increasing: warming atmospheric temperatures increases evaporation, producing more rain and sea-level rise along the coasts. One study by the National Science Foundation found that extreme rainfall events in the US could become three times more likely and increase up to 20% in severity within the next 45 years.¹³

Floods, as defined by the Natural Resource Defense Council, are the accumulation of water in normally dry land caused typically by an overflow of inland (rivers and streams) or coastal waters or by an unusual accumulation of water from prolonged rains, storm surges or

¹² Elisabetta Motta, "The local dimension in the degrowth literature. A critical discussion," *Journal of Political Ideologies* 25, no. 1 (2020): 78-93.

¹³ National Science Foundation, "Extreme Rainfall Projected to Become More Severe, Frequent With Warming," NSF - U.S. National Science Foundation, June 14, 2022, <https://www.nsf.gov/news/extreme-rainfall-projected-become-more-severe>.

sudden snowmelt.¹⁴ Often the management of waterways (encasing within concrete) and alterations to land play a large role in the risks of flooding. Additionally, increased urbanization creates impermeable surfaces diverting natural drainage systems and driving worsening floods. The action of daylighting would mitigate this threat through the creation of porous terrains that allow absorption lowering the risk of high flowing water.

5. Daylighting

Daylighting is the process of exposing some or all of a previously covered river, stormwater drain, or stream. As the environmental advocacy organisation American Rivers explains, this process can exist in several forms:

1. Natural restoration: restoring a stream to “natural” stream conditions.*
2. Architectural restoration: restoring a stream to open air, flowing water but within a constructed channel.
3. Cultural restoration: celebration of a buried stream through markers or public art used to inform the public of the historic stream path, although the stream remains buried.

* The American Rivers organization makes note that natural restoration can often appear in “constructed” channels but rather than concrete lined as seen in architectural restoration, these channels are human attempts at recreating the natural systems of a porous and irregular streambed.¹⁵

II. Historical and theoretical context

A. *Capitalism and Sustainability*

The Earth has a finite amount of resources, making endless economic growth impossible. The goal of capitalism is to generate profit and accumulate capital for private individuals. Nothing about capitalism encourages people to consider the well-being of anything besides themselves and their ability to make money. Yet, the accumulation of wealth must come from somewhere. That "somewhere" comes from concepts of private property, extractive labor

¹⁴ Melissa Denchak, “Flooding and Climate Change: Everything You Need to Know,” Natural Resources Defense Council, November 3, 2023, <https://www.nrdc.org/stories/flooding-and-climate-change-everything-you-need-know>.

¹⁵ Nicholas A. Tonelli and American Rivers. “Daylighting Streams: Breathing Life Into Urban Streams and Communities,” Report. American Rivers, n.d, https://www.americanrivers.org/wp-content/uploads/2016/05/AmericanRivers_daylighting-streams-report.pdf, 8.

processes and the destruction of both urban and rural places. Therefore, for the sake of the planet and people alike, we hope to transition away from America's current insatiable desire to endlessly accumulate more wealth with no end in sight. We believe that degrowth offers a tangible first step in focusing on the needs of people and the planet rather than the unsustainable desires of corporations and extremely wealthy individuals.

B. Racial Capitalism

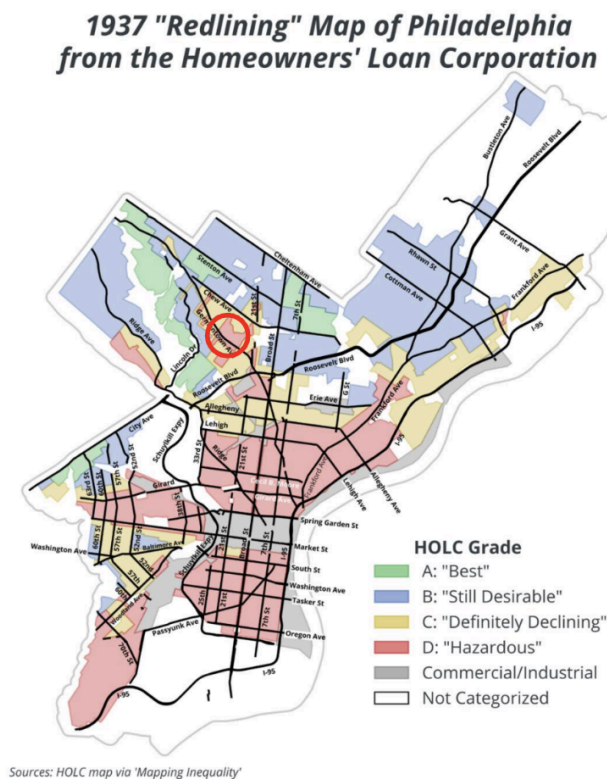


Fig. 2 1937 Redlining Map of Philadelphia with Germantown Neighborhood Circled

To fully comprehend capitalism, as many critics have argued, one must consider the history of race relations in the United States. The success of capitalism has been made possible through the creation and maintenance of racial divides. The theory of racial capitalism conceptualizes how modern society is a product of racism and capitalism through valuing

products, places, and people based on racial differences.¹⁶ Therefore, when we consider the socio-political-economic implications on the lived environment of those in Germantown, we must discuss the role of America's racial stratifications.

For example, during the first half of the 1900s, the US local and federal governments worked to maintain racial divides through mapping out neighborhoods and valuing areas on a scale of financial riskiness. The discriminatory process of drawing lines between areas, called redlining, was used to maintain white wealth by making it more difficult for Black and Brown individuals to become homeowners. In the Philadelphia city controller's document on structural racism, Rhynhart explains that the city is one of the most racially segregated cities due to the process of redlining. In *fig. 2*, we can observe how large areas of the city were categorized as hazardous while others were deemed desirable. The areas devalued the most were typically composed of "African Americans, immigrants, and Jews".¹⁷ We have added a circle in red on this map to highlight how Germantown, a historically Black neighborhood, was largely devalued. Due to structural economic processes such as redlining, it is vital to recognize capitalism as more than an economic institution, but one that disproportionately disadvantages people while uplifting a select few.

The consequences of racist, predatory, and capitalistic policies such as redlining have been shown to "have lasting impacts on hazard vulnerability today, leaving geographic areas within Philadelphia at higher risk to hazards such as heat, flooding, and gun violence".¹⁸ Racism, poverty, and environmental risk in Philadelphia are linked together. It becomes critical to consider how ecological impacts such as flooding are not mere coincidences, but rather are often systemically created and structurally reinforced. With this context, it becomes clear that current economic and political interventions can hinder the necessary change needed to keep communities healthy. There is a clear need to reconsider the capitalist drive within America for the betterment of our everyday lives.

¹⁶ Zawadi Rucks-Ahidian, "Theorizing Gentrification as a Process of Racial Capitalism," *City & Community* 21, no. 3 (November 15, 2021): 173–92, <https://doi.org/10.1177/15356841211054790>, 178.

¹⁷ City of Philadelphia Office of the Controller, "Mapping the Impact of Structural Racism in Philadelphia," 2019, 3.

¹⁸ City of Philadelphia Office of Emergency Management, "2022 All-Hazard Mitigation Plan," 2022, 50.

C. Gentrification in Germantown

Germantown community members, Reverend Chester Williams and Daneen Brockington, have noticed in recent years an influx of buildings being developed. The transformation of infrastructure rightfully calls for asking if the neighborhood is being gentrified. Gentrification can be defined as “the socioeconomic upgrading of a previously low-income central city neighborhood, characterized by the influx of residents of a higher socioeconomic status relative to incumbent residents and rising home values and rents.”¹⁹ Although we are uncertain if Germantown is being gentrified, it must be considered based on the marketing of the neighborhood, programs like Jumpstart, and observations from community members.

If capitalism cannot be divorced from racial ideologies, then we must consider the role of race in Germantown as it relates to gentrification. On one hand, majority Black and Brown neighborhoods have often been devalued compared to similar economically stratified neighborhoods.²⁰ However, as Rucks-Ahidiana points out, Black and Brown neighborhoods might become highly valuable if they are in proximity to city centers and [through] a commodification of diversity and culture.²¹ It seems possible that Germantown has begun to be marketed as a higher-value neighborhood, making it “worthy” of gentrification. Take, for example, the following text from the city's website:

*“Looking to experience Philly history off the beaten path? Germantown’s got you. Tucked between Hunting Park and Mt. Airy, Germantown is one of the city’s oldest settlements, with 300-year-old homes, wide cobblestone streets, surprising museums, spectacular festivals and lush green spaces. Deeply tied to Black history and once a key stop on the Underground Railroad, Germantown continues its legacy of equality with a thriving community of Black-owned businesses. Just 20 minutes from Center City, this hidden gem stretches along Germantown Avenue and offers a relaxed, history-rich complement to the well-known historic sites in Old City.”*²²

¹⁹ Lei Ding, Jackelyn Hwang, and Eileen Divringi, “Gentrification and Residential Mobility in Philadelphia,” Working Paper (Federal Reserve Bank of Philadelphia), September 22, 2016, <https://doi.org/10.21799/frbp.wp.2015.36>, 38.

²⁰ Zawadi Rucks-Ahidiana, 174.

²¹ Ibid., 179.

²² “A Guide to Germantown: Philly’s Hidden Historic Treasure,” Visit Philadelphia, December 1, 2025, <https://www.visitphilly.com/areas/philadelphia-neighborhoods/germantown/>.

The culture, green spaces, and “thriving community” are something which those living in Germantown have struggled and continue to struggle to maintain because of ongoing issues with flooding and dismissal from government officials. Throughout Philadelphia, one can observe hundreds of community gardens, which are often cultivated on vacant lots. Yet, without legal ownership over lots, spaces where people come together to grow food, are subject to development.²³ In Germantown, places such as Nicholson Church Lane Community Garden, originally were threatened by development, but through continuous effort, have been maintained to provide a space for beautiful connection. Vacant lots hold potential for positive change, including ecological restoration and community engagement. At the same time, they are subject to economic whims by outside developers who are unaware of community needs.

Throughout Germantown, one will walk past numerous vacant lots and uninhabitable buildings. These spaces hold the potential to create positive and negative impacts on the community. Projects such as Nicholson Church Garden and Awbury Arboretum demonstrate how land can be used to serve both the community's needs and the environment. However, properties are subject to the desires of private developers rather than the needs of the community. For example, in recent years, Germantown High School, which was shut down in 2013, has been converted into an apartment complex called “The Annex”. The Annex is promoted as a set of luxury apartments “seamlessly transformed from a historic school building in the vibrant Germantown neighborhood.... [with] a fusion of history and contemporary elegance.”²⁴ Studio and 1-bedroom apartments range from \$1,050 to \$1,425, which is comparable to the median gross rent in East Germantown (\$1,169).²⁵ Although the prices of these apartments are comparable to those of other apartments in the area, that does not necessarily mean they are affordable to those within the community. The median household income in Germantown is around \$53,161, which is two-thirds of the average \$77,971 in Pennsylvania. This translates to around 24.2% of people in Germantown living below the poverty line, which is more than

²³ “History,” Neighborhood Gardens Trust, accessed April 27, 2026, <https://www.ngtrust.org/history/>.

²⁴ Inc. Zillow, “The Annex Apartments - 1518 Spruce St Philadelphia Pa,” Zillow, accessed April 27, 2026, <https://www.zillow.com/apartments/philadelphia-pa/the-annex-apartments/98vDCh/>.

²⁵ U.S. Census Bureau, “Housing,” 2024 American Community Survey 5-Year Estimates, in ZCTA5 19138, accessed [April 2026], https://data.census.gov/profile/ZCTA5_19138?g=860XX00US19138#housing.

double the rate of the average 11.7% in Pennsylvania.²⁶ What Germantown needs then is not luxury apartments but rather affordable, safe, and environmentally sustainable housing options.



Fig. 3 Development of the Annex on the site of the old Germantown Highschool

We do not intend to single out the developers of the Annex or any other developer for that matter; instead, we hope to illuminate a larger problem across Germantown, Philadelphia, and America as a whole. Within a capitalist economy, people's needs, such as housing, are dependent on choices that private developers make. It is in the interest of private developers to make the most amount of money possible rather than considering the necessities of people, especially those who are already impoverished. What people need are not more expensive homes but rather safe, healthy, affordable, and environmentally sustainable housing.

²⁶ U.S. Census Bureau, "Income and Poverty," 2024 American Community Survey 5-Year Estimates, in ZCTA5 19138, accessed [April 2026], https://data.census.gov/profile/ZCTA5_19138?g=860XX00US19138#income-and-poverty.

However, this issue extends beyond large apartment developers but can apply to the phenomenon of “house-flipping.” Jumpstart Germantown is a program working to teach people in the community skills surrounding financial literacy for urban development.²⁷ With Jumpstart’s workshops, people can learn how to purchase, renovate, and sell properties. After going through their program, people are eligible to receive loans offered by Jumpstart. On the one hand, it seems preferable for the development of properties within Germantown to be done by community members themselves rather than by outsiders. Theoretically, if homes are being developed by those of a community, one would expect them to implement changes that would reflect the values and needs of those who live there. For example, perhaps those from Germantown who flip homes within the neighborhood would choose to implement flood prevention measures. However, even a program such as Jumpstart exists for the purpose of creating profit rather than directly catering to those in need. When the focus is on profit, the risk of gentrification becomes a concern.

What will happen to the people in Germantown if the neighborhood gentrifies? In a 2020 study looking at credit information from over 50,000 random individuals from Philadelphia, researchers found that those living in neighborhoods of lower economic status were not necessarily more likely to move than those in non-gentrifying neighborhoods. At the same time, those of lower economic status *were* more likely to move to neighborhoods of lower economic status. However, a limitation of this study is the fact that it relies on a sample of people who have credit information accessible. Therefore, it can’t consider individuals who live on the fringes, such as those who may not be paying taxes. That being said, the authors do acknowledge that “gentrification’s consequences on the residential mobility patterns of financially disadvantaged residents are unequally distributed by the racial contexts in which gentrification is taking place.”²⁸ There are impacts caused by gentrification across socioeconomic inequalities, and these differences are further fragmented by racial categories. Based on this study, if Germantown continues on a trajectory of typical gentrification, people who can no longer afford to live there will be pushed to move into communities of lower socio-economic status, and those most impacted will be Black and Brown residents.

²⁷ Jumpstart Germantown, accessed April 27, 2026, <https://www.jumpstartgermantown.com/>.

²⁸ Jackelyn Hwang and Lei Ding, “Unequal Displacement: Gentrification, Racial Stratification, and Residential Destinations in Philadelphia,” *American Journal of Sociology* 126, no. 2 (September 1, 2020): 354–406, <https://doi.org/10.1086/711015>, 354.

D. *Zoning issues*

1. Zoning in Germantown

Zoning laws exist to categorize what is allowed to happen on a section of land (Philadelphia Zoning).²⁹ On a broader scale, this requires some land to be used for housing while other land is zoned for retail use. Areas zoned for housing purposes have sub-categories used for smaller differentiations, such as a single-family unit vs. an apartment complex. One additional method used to dictate what can and cannot be done in an area is overlay laws.³⁰ For example, if there is an area that has ecological concerns, overlays can be used to require conservation and environmental actions to prevent further harm. The way land-use is categorized has the potential to guide land use as one which protects the needs of people and the planet.

In Germantown, zoning has not been updated for more than four decades, indicating a clear need for zoning updates.³¹ The Philadelphia City Planning Commission has created a 2035 plan to address the needed reworking of zoning areas within Germantown, in addition to rethinking structural infrastructure throughout the city. Although many homes have basements that flood, the only inclusion of flood prevention measures in the 2035 plan is to limit housing construction along Belfield Avenue. Yet, flooding is not confined to this street. When given the opportunity to provide feedback for the 2035 plan, community members commented on the need for greater flood prevention updates. For example, one community member commented that “PWD [Philadelphia Water Department] has identified 24 flood-prone intersections and 2,700+ basements with flood risk in Historic Wingohocking Creek Watershed. Plan's limited focus on 3 parcels does not do anything to promote Wingohocking flooding solution and puts a band-aid on the problem. We need a Watershed-wide overlay district, not a quick fix for a few sites.”³²

²⁹ City of Philadelphia Planning and Development, “Philadelphia Zoning Code Quick Guide,” 2022, 5.

³⁰ Pennsylvania Department of Community & Economic Development, “Overlay Zoning,” 2019, 1.

³¹ Admin, “Rezoning Northern Germantown,” phila2035, July 6, 2021, <https://www.phila2035.org/single-post/rezoning-northern-germantown#:~:text=PCPC%20is%20working%20to%20update.Northwest%20District%20Plan%20in%202018.>

³² Philadelphia City Planning Commission, “Germantown North Zoning Remapping Open House Comments,” 2021, 4.

2. By-right

When talking to Reverend Chester Williams, he repeatedly emphasized by-right zoning laws as a pressing concern in Germantown. By-right allows developers to buy property and request to make changes to a parcel of land with much quicker permit issuance (around 1-2 months) without needing the approval of the community.³³ Reverend explains that when outside developers come in, they can update properties through by-right and “don’t care about flooding” or other needs of the community. This is allowed to happen because the changes made to the land fall within the existing zoning regulations and codes. However, since zoning laws have not been largely updated in over 40 years, what developers are allowed to change on properties may not reflect the concerns and needs of today. This is even more of a concern with outside developers who are focused on profit and unaware of the needs of the community. Through by-right, can quickly develop properties without needing to consider environmental or social needs such as flooding risk. This issue calls for greater flood legislation to be in place beyond the reduction of construction on Belfield Ave.

3. Alternative future

Initially inspired by the Reverend’s adamant condemnation of by-right zoning in Philadelphia, and from additional conversations with community members through the Watershed about new developments without adequate flood-protection, home-flipping, and gentrification, it became clear to us that zoning is an integral chapter within the story of flooding in Germantown. As discussed previously, zoning policies clearly do not reflect the priorities and needs of Germantown residents, particularly as it pertains to flooding issues. But is there any alternative? Even within our own research, we found ourselves bogged down and discouraged by the intimidating nature of zoning as a bureaucratic, confusing territory within city policy. Advocating for new zoning practices seems an impossible task, especially within an urban context where zoning is intensely entrenched in both public government and private ownership interests.

However, just as degrowth bravely asserts a radical alternative to the capitalist, growth-centered systems of the world, radical reimaginations of zoning offer the potential to

³³ Philly Zoning Team, “By-Right Development,” Philadelphia Zoning, February 24, 2018, <https://philadelphiazoning.com/2018/02/by-right-development/>.

rethink a system that clearly isn't working to support the well-being and needs of city residents. There are three planning mechanisms that maintain economic competition in contemporary urban development: the organization of city regions as competitive land markets, the maintenance of land scarcity to generate competition, and the institution of Euclidean land zoning. Euclidean zoning is responsible for the traditional development patterns within the US built environment, generating suburban sprawl, racial and socioeconomic segregation, and environmental costs.³⁴ Essentially, Euclidean zoning assumes the city functions best when land is segregated into distinctly separate, single-use zones.³⁵ Since the mid 1970s, planners and urban activists have criticized Euclidean zoning's effects on the ability of communities to both self-govern and innovate their living practices as needed.³⁶ Euclidean zoning works to make land-use within cities legible to government city planners; zoning and the commodification of land depends on its "possible future uses, not its actual realized use."³⁷ Ultimately, Savini argues not for the abolition of zoning but rather to "reimagine normative conceptions of 'good land use,' beyond property value."³⁸ Savini proposes "habitability" as an alternative to Euclidean-style zoning practices. Habitability prioritizes basic living rights over a right to property and exchange. Savini further outlines how examples from Indigenous land management practices may guide reimagination of standard American urban planning practices, centralizing the social-ecology of an area within the planning process through identifying, maintaining and restoring the social and ecological qualities of an area.³⁹

Whether Philadelphia abandons Euclidean-style zoning for habitability-based urban planning, understanding the ways in which traditional zoning work against adaptable zoning which supports the needs of both local communities and local environments is crucial to understanding why zoning and by-right is such a central problem within the story of flooding in Germantown. Within the rigid structure of zoning policy, the possibility of daylighting abandoned lots under which the historic streams of Philadelphia run is made difficult due to the bureaucratic red-tape of zoning—despite the fact that daylighting these streams would address

³⁴ Eliza Hall, "Divide and sprawl, decline and fall: A comparative critique of Euclidean zoning," *U. Pitt. L. Rev.* 68 (2006): 916.

³⁵ *Ibid.*, 918.

³⁶ Federico Savini, "Towards an urban degrowth: Habitability, finity and polycentric autonomism," *Environment and Planning A: Economy and Space* 53, no. 5 (2021): 1084.

³⁷ *Ibid.*, 1084.

³⁸ *Ibid.*, 1085.

³⁹ *Ibid.*, 1089.

the needs of the water to flow as it wants to and the needs of residents struggling against flooding issues.

III. Proposal

A. Social transition

1. Mutual Aid System

As described in the introduction, mutual aid is the reciprocal exchange of resources and services amongst community members to meet immediate needs and build collective power to bypass the systems that aim to keep people marginalized. Rather than charity's structure of hierarchical giving under the assumption of one party having more power and agency over the other, mutual aid recognizes the world as a network—when a piece of that system is not cared for, the entire structure threatens to collapse.

The term mutual aid was coined by the Russian anarchist Peter Kropotkin in a series of essays countering the theories of social darwinism. Kropotkin posits the theory asking, who are the “fittest” within the claim “survival of the fittest” and concludes the most successful are the ones who mutually support their communities for the prosperity of the whole.⁴⁰

Mutual aid as an alternative structure is not new. Robin Wall-Kimmer, in her book *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge and the Teachings of Plants*, describes indigenous communities in the Americas utilizing collectivism and reciprocity, “What happens to one happens to us all. We can starve together or feast together. All flourishing is mutual...” (pg 15).⁴¹ In 1969, the Black Panther Party began their free breakfast program for children understanding that directly addressing the issues of hunger, class oppression and racism would build trust in the movement and increase community welfare.⁴²

In response to conversations with members of the Germantown community on the topic of flooding risks and decaying infrastructure, we propose the formalization of a new mutual aid network, aimed at lessening the burden from the individual and onto the

⁴⁰ Peter Kropotkin, *Mutual Aid: A Factor of Evolution*, New York: McClure Phillips & Co., 1902, <https://theanarchistlibrary.org/library/petr-kropotkin-mutual-aid-a-factor-of-evolution>.

⁴¹ Robin Wall Kimmerer, *Braiding Sweetgrass*. Milkweed Editions, 2013, <https://ccncsj.ca/wp-content/uploads/2025/01/Braiding-Sweetgrass-Robin-Wall-Kimmerer.pdf>.

⁴² Mary Potorti, “‘Feeding the Revolution’: The Black Panther Party, Hunger, and Community Survival,” *Journal of African American Studies* 21, no. 1 (March 1, 2017): 85–110. <https://doi.org/10.1007/s12111-017-9345-9>.

collective. This sentiment was echoed in conversations with Germantown resident and local activist, Reverend Chester Williams. The Reverend explained many Germantown residents feel under-represented and ignored by their city council members and the state;

“There is no council person and state rep that is connected with this district that does not know... but for some reason behind closed door meetings discussing flooding with [the exception of] maybe someone from the water department [there is no one] on the zoning board. This is critical...[we] mention it all the time but they come back from the meetings and like you just found out, nothing was done. The public don’t know about those lobby meetings... and if they couldn’t come they should be pushed to put it on zoom.”

The collective voice of those actually affected by the flooding is necessary for understanding the pain and hardship faced by the community and would correctly address the needs of those most at risk. As proposed by the Reverend, meeting these needs would involve such as a free program to fix basement flooding, reallocation of money from the state to updated infrastructure, and a neighborhood permanent seat on the Water Department council—as proposed by the Reverend.

In this gap left by the inaction of the city, mutual aid stands as a form of immediate support for the neighborhood. **We propose the creation of a resource library and skills exchange run and circulated by local residents and community organizations who opt into the program.** Through the community space of the WaterShed, the resource library would appear as an organized collection of informational materials, and tools available to the public, in defense of flooding. Tools, such as plywood to raise appliances off ground levels, wet-dry vacuums, and portable submersible pumps; that lower the threat of high cost damage in cases of residential flooding. The skills exchange similarly would rely on public labor to trade skillsets amongst community members to both aid in:

- ❖ **flood prevention:** help applying for flood homeowners insurance or building the raised platforms for electrical appliances,
- ❖ **flood management:** installing french drains,
- ❖ **flood support:** restoring damaged basements and sourcing/replacing lost valuables.

Both programs enact the core principles of mutual aid; community organizing based on reciprocity, where participants share resources, skills, and care to meet and uplift the needs of others.

Although incredibly helpful, we do want to acknowledge that mutual aid is also often forced upon communities as a result of the destructive system of capitalism. Low wages and gaps in government support leave people reliant on mutual aid, removing accountability from the mega corporations that benefit from keeping these populations underpaid. While mutual aid stands as a strong solution to immediately meet the needs of communities, true support comes from legislation that prevents these corporations from stifling the populations.

2. Circular Economy in Partnership with RAIR

RAIR is an artist-in-residence program based at Revolutionary Recovery, focusing on using materials from the waste stream. Revolutionary Recovery's business model is based on accepting construction and demolition waste (C&D) and finding ways to utilize and/or recycle the materials before sending them to landfills. Although RAIR is focused on the arts, in 2020, they launched a "circular economy" pilot project to bridge material gaps within the Philly community. Largely, they pull dimensional lumber from the waste stream and freely offer these materials to community farms, which are able to then grow more food for the communities they serve. For example, Glover Gardens has been able to distribute over 60 raised garden beds to communities in Southwest and North Philadelphia who now have access to fresh produce.⁴³ This example demonstrates how mutual-aid efforts have a multi-layered impact. Mutual aid can decrease waste and increase community connection, offering both physical and mental health benefits.

Capitalism functions through alienating people from their means of subsistence, such as making it difficult for people to grow their own food. Rather, it pushes communities to rely on a wage to access life-sustaining materials like food, housing, and health care. Yet, the means of subsistence offered, especially to those of lower socio-economic status, are of lesser quality, causing many health issues today, such as cancer. Alternative economies offer an avenue towards a healthier, happier, and more sustainable future.

⁴³ RAIR. "RAIR," n.d. <https://www.rairphilly.org/>.

B. Physical systemic infrastructure shifts

1. Daylighting

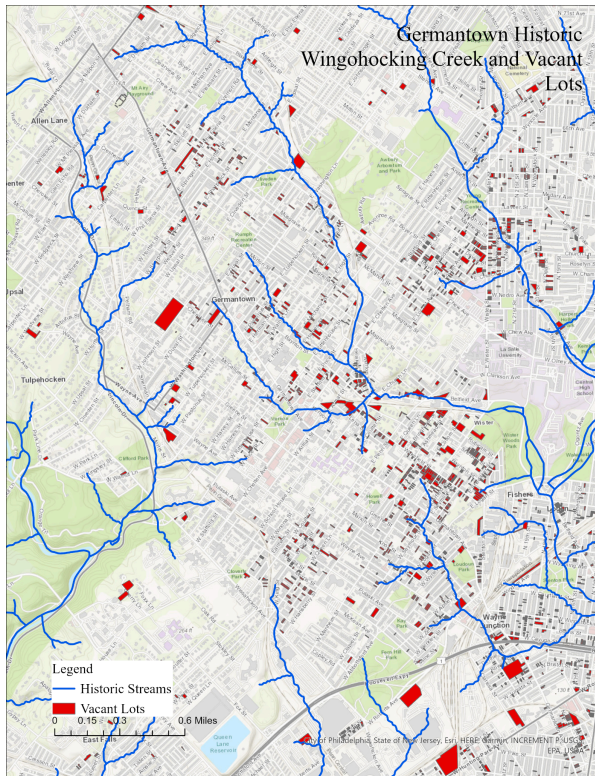


Fig. 4 Historic Philadelphia Rivers and Vacant Lot

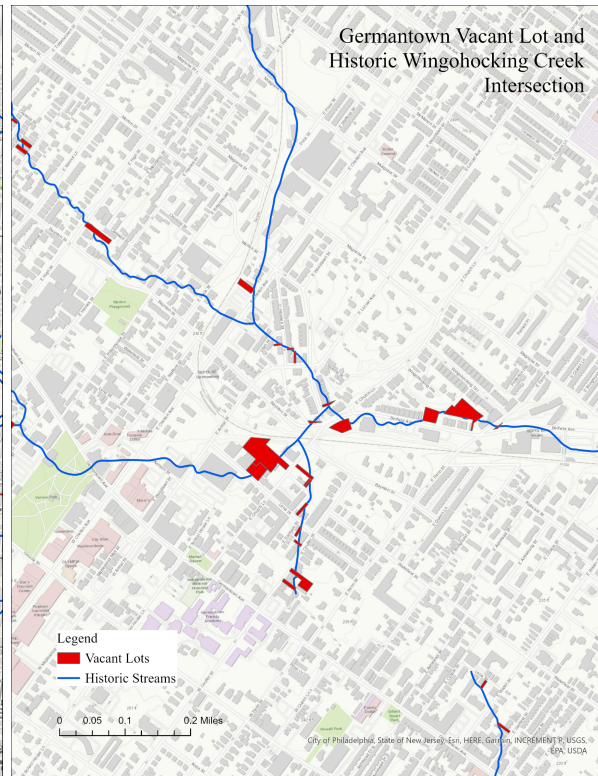


Fig. 5 Intersection of Wingohocking and Vacant Lots in Germantown

Within the central boundaries of the Germantown neighborhood, sits around 30 vacant lots intersecting with the historical pathways of Philadelphia's largest covered rivers (*fig. 5*), the Wingohocking. This data was pulled from the *Philadelphia's Vacant Property Indicators Model*, “an analytical model developed by the City of Philadelphia Office of Innovation and Technology in cooperation with Department of Licenses and Inspections, Office of Property Assessment, Philadelphia Land Bank and Philadelphia Water Department,” and mapped using ArcGIS to find the overlaps (*figs. 3 and 4*).⁴⁴ These lots are of mixed ownership; city land and private landlords,

⁴⁴ City of Philadelphia Office of Innovation and Technology, Department of Licenses and Inspections, Office of Property Assessment, Philadelphia Land Bank, and Philadelphia Water Department, “Philadelphia’s Vacant Property Indicators Model,” Data set. City of Philadelphia, 2026, <https://www.arcgis.com/apps/webappviewer/index.html?id=64ac160773d04952bc17ad895cc00680>.

which leaves opportunities for local government and developers to invest into the local community.

What would it look like to unearth these streams and transform the abandoned lots into something of beauty and community care? Daylighting exists as a two-pronged solution, providing both social revitalization and environmental reparations. In reference to the ideas on development explored in the sections above, we hope to explore a speculative proposal for a green infrastructure transformation in the Germantown neighborhood through the process of daylighting vacant lots to ask the question, *what does it mean for water to return?*

Daylighting is not new to Philadelphia. In fact, the Awbury Arboretum located in Germantown, daylit a small portion of the Winghocking Creek as part of a larger restoration project aided by the Philadelphia Water Department, Pennsylvania Department of Environmental Protection, and others. The Wetland Restoration is projected to alleviate the stormwater flooding that regularly occurs in the Washington Lane and Chew Avenue area. In addition to lessening flood impacts, the arboretum found drawing the stormwater from Washington Lane to the restored wetlands reduces the amount of pollutants that reaches the city sewer system from runoff.⁴⁵



Daylighting and these types of floodplain revitalization increase the hydrologic capacity. Whereas “flooding... is a direct result of a loss of water absorption capacity within the landscape,” daylighting creates a region of permeable surfaces.⁴⁶ Whereas, “the shape or

⁴⁵ Branda O’Neil and Branda O’Neil, “Wetlands Restoration Is an Environmental Success Story,” Awbury Arboretum | Philadelphia, PA, May 20, 202, <https://awbury.org/wetlands-article/>.

⁴⁶ Nicholas A. Tonelli and American Rivers, 6.

morphology of the channel also impacts downstream flooding rates... with development, straightening, deepening, or piping small streams alter[ing] ecological function by making the hydrology smoother, essentially transmitting the flood downstream faster,” daylighting's reduction in channelization through rough streambed, woody debris, and irregularly shaped corridors, allows for water to be slowed in the event of heavy rains.⁴⁷ And similarly reduces chokepoints where water can back up, lowering the risk of localized flooding.

The burial of streams and extreme alterations to the natural land, prove to be an encompassing issue. The impervious surfaces and fast runoff volumes that cause flooding, also decrease the ability of groundwater recharge. Furthermore, the inability for water to access these historic headwater streams and flow through buffer zones results in a lack of filtration from pollutants and fertilizers. Under normal circumstances, “These small streams and their riparian corridors absorb pollutants via uptake by in-stream and surrounding riparian vegetation, algal uptake, and microbial uptake resulting in less water pollution released into stream flows.”⁴⁸ The loss of these natural processes degrades water quality, and ruins the habitats of varying species, necessitating a larger need for greenspace filtration. The protection of these streams is vital for the continuation and preservation of the larger ecosystem, of which the Germantown community is part of.

Yet the damages of these actions go beyond environmental destruction. The cost of flooding damage in the United States annually totals between \$179.8 billion and \$496 billion when factoring in a broader spectrum of impacts: annual loss in tax revenue, increases in insurance premiums, damage to ecosystem services, and damage to transit infrastructure.⁴⁹

In Philadelphia the cost of flooding damage is often left on the residents. Homeowners insurance does not cover flood damage and while the US government does provide flood insurance through the National Flood Insurance Program, managed by FEMA (Federal Emergency Management Agency), properties outside of FEMA designated flood zones could be subject to specific insurance requirements in order to obtain a mortgage.⁵⁰ In the 2024 FEMA

⁴⁷ Ibid., 6.

⁴⁸ Ibid., 7.

⁴⁹ United States Joint Economic Committee, “Flooding Costs the U.S. Between \$179.8 and \$496.0 Billion Each Year,” Flooding Costs the U.S. Between \$179.8 and \$496.0 Billion Each Year - United States Joint Economic Committee, June 10, 2024, <https://www.jec.senate.gov/public/index.cfm/democrats/2024/6/flooding-costs-the-u-s-between-179-8-and-496-0-billion-each-year>.

⁵⁰ FEMA. “Flood Insurance,” January 2, 2026. <https://www.fema.gov/flood-insurance>.

flood risk map of Philadelphia the only areas highlighted as risky were along the banks of the Schuylkill and Delaware Rivers, ignoring the very present history of flooding further inland. This blindspot in the research presents a dangerous oversight in federal aid and individual costs of flooding management can reach up to the thousands.⁵¹ One resident spoke in a newspaper article explaining when her sewage line backed up in 2023, the bill for clearing totaled nearly \$23,000. While the case of Linda Bell could be considered lucky, insurance eventually covered around 80% of the cost, her premium still went up by a dramatic \$531.40.⁵²

2. Cost of daylighting

Given overwhelming impacts of encased waterways, daylighting appears as the singular viable solution. But this is not without risks. It must be noted, daylighting usually does include a larger upfront price. The American Rivers organization calculated the general price of daylight to an estimation of around \$1,000 per linear foot of stream unearthed. However, within the case studies they provided, cost values ranged from \$15 to \$5,000 per linear foot. Similarly noted, the cost was shown to increase or decrease depending on the number of factors included; like the extent of urbanization on the proposed land, the payment of workers vs. volunteers, and the addition of community amenities once the project was complete.⁵³

Yet in an article published by the Smithsonian, researchers found the process of daylighting in multiple urban centers across the US led to a renewal for the cities both economically, as in the case for Napa's economic growth of 1 million, and a release of pressure on overworked stormwater and sewer systems. The article concedes, while the projects can be costly, in many scenarios returning to nature was cheaper than maintaining yearly flooding protection.⁵⁴ The white paper by American Rivers continues, "Daylighting... provides economic benefits to communities through cost effective alternatives to ongoing culvert maintenance and

⁵¹ "FEMA Flood Map Service Center | Search by Address," n.d.
<https://msc.fema.gov/portal/search?AddressQuery=Philadelphia>.

⁵² Carla Robinson, "Fixing the Chronic Flooding in Northwest Philadelphia Will Be a Major Undertaking. In the Meantime, Residents Live With the Danger and Expense," Grid, March 28, 2024,
<https://gridphilly.com/blog-home/2024/03/28/fixing-the-chronic-flooding-in-northwest-philadelphia-will-be-a-major-undertaking-in-the-meantime-residents-live-with-the-danger-and-expense/>.

⁵³ Tonelli, Nicholas A. and American Rivers., 16.

⁵⁴ Jim Morrison, "How 'Daylighting' Buried Waterways Is Revitalizing Cities Across America," Smithsonian Magazine, March 15, 2023,
<https://www.smithsonianmag.com/innovation/how-daylighting-buried-waterways-is-revitalizing-cities-across-america-180981793/>.

by keeping stormwater out of combined sewer systems, thereby reducing water treatment costs.”⁵⁵

We took these estimates of cost into consideration and proposed daylighting only within areas of Germantown that are already undeveloped or unused. What we found was many of the vacant lots of Germantown are concentrated along the same route of the historic creek. In the maps provided (*figs. 4 and 5*), the cross streets of Church Ln and Belfield Ave, show a large cluster of vacant lots. This area of Germantown is one of the most frequently flooded regions in the neighborhood as shown in the flood map located at WaterShed (*fig. 6*). Uncovering these branches from the Wingohocking that intersect with already available lots would provide a unique opportunity for a green infrastructure initiative that is more cost effective than yearly flooding repairs.

Additionally, Philadelphia is currently spending billions of dollars on its 25 year *Green City, Clean Waters* plan, which officially began in 2011.⁵⁶ The Philadelphia Water Department (PWD) originally estimated the cost of the plan to be about \$2.4 billion in investments citywide, with \$1.7 billion going to green infrastructure, and hundreds of millions going to upgrade Philadelphia’s three wastewater treatment plants and reline pipes, and \$420 million set aside for “flexible” spending on green or concrete infrastructure as needed.⁵⁷ However, this bill has ballooned to at least \$4.5 billion at its halfway point, with costs borne by the designated “stormwater charge” tacked on to Philadelphia residents’ monthly water bills.⁵⁸ Within the context of this massive spending plan, daylighting streams would be a relatively reasonable price tag for a real, at-the-root solution to flooding in Germantown—and possibly throughout Philadelphia. By allowing these buried waterways to breathe, the city would be spending money on a long-term solution to the historic issue that is responsible for the flooding issues faced in Germantown.

⁵⁵ Nicholas A. Tonelli and American Rivers, 1.

⁵⁶ Lauren Sell, “Green City, Clean Waters,” March 4, 2026, <https://water.phila.gov/green-city/>.

⁵⁷ Kyle Bagenstose, “Philly Is Spending \$2B+ to Fix Its Sewage Problem. But Is Green City, Clean Waters Working?” Grid Magazine, February 1, 2024, <https://gridphilly.com/blog-home/2023/12/01/philly-is-spending-2b-to-fix-its-sewage-problem-but-is-green-city-clean-waters-working/>.

⁵⁸ Circle of Blue, “Billions Over Budget: Questions Swirl Over Philly’s ‘Green’ Fix for Sewers,” *Circle of Blue*, March 5, 2025, <https://www.circleofblue.org/2024/supply/water-management/billions-over-budget-questions-swirl-over-phillys-green-fix-for-sewers/>.



Fig. 6 Area of concentrated flooding and location of converging river and vacant lots

We can approach daylighting like an exercise in mutual aid: what can we give to the earth, and what will it give back? We repair the stream, and the water gives back protection from flooding. We revitalize the greenery and get back community engagement and economic vitality. The reciprocal relation between humanity or the urban landscape and the earth is one we should cultivate and the process of daylighting is a path forward. Daylighting then becomes the process of building relationships with the land in opposition to the capitalistic structures that drive profit over people.

3. Bio swales

The daylighting of Philadelphia's historic streams would not only *allow the water to return*, but also presents the opportunity for greenspace additions to Germantown in the form of sponge parks, rain gardens, and bio-swales and for natural filtration of polluted run-off stormwater. While slightly different, these techniques are essentially designed to mitigate flooding and stormwater run-off pollution in urban environments. Bioswales are particularly promising in partnership with daylighted streams, as they take the form of planted trenches.⁵⁹ The beneficial impacts of bioswales include significantly reducing runoff, lowering

⁵⁹ Emine Şen and Beyza Şat, "Bioswales as engineering functions of nature-based solutions to increase urban resilience: A review," *Forestist* 74, no. 2 (2024): 253.

contamination through soil filtration, and easing energy burdens on stormwater treatment infrastructure.⁶⁰

The Gowanus Canal Sponge Park offers a successful example of a multifunctional public green space that “slows, absorbs, and filters surface water runoff with the goals of remediating contaminated water, reducing stormwater runoff volumes, creating habitat, and providing open space and environmental education opportunities to underserved neighborhoods.”⁶¹

By planting native plants that thrive and work best in rain gardens, greenspaces fostered around these daylighted sections of Philadelphia’s historic streams would magnify their ability to mitigate flooding issues in Germantown. Janet Marinelli, the former director of publishing at the Brooklyn Botanic Garden, compiles a comprehensive list of native plants suitable for rain gardens in the Northeast:

<u>Northeast Native Rain Garden Plants⁶²</u>	
Wildflowers, Ferns, Grasses, and Sedges:	Trees and Shrubs:
<i>Asclepias incarnata</i> , Swamp milkweed	<i>Amelanchier laevis</i> , Shadbush
<i>Chelone glabra</i> , White turtlehead	<i>Asimina triloba</i> , Pawpaw
<i>Eupatorium maculatum</i> , Joe-pye weed	<i>Betula nigra</i> , River birch
<i>Lobelia cardinalis</i> , Cardinal flower	<i>Cephalanthus occidentalis</i> , Buttonbush
<i>Lobelia syphilitica</i> , Blue lobelia	<i>Clethra alnifolia</i> , Sweet pepperbush
<i>Monarda didyma</i> , Oswego tea	<i>Cornus amomum</i> , Silky dogwood
<i>Vernonia noveboracensis</i> , Common ironweed	<i>Fothergilla gardenii</i> , Dwarf fothergilla

⁶⁰ Ibid., 254.

⁶¹ Susannah C. Drake and Yong Kim, "Gowanus Canal Sponge Park™," *Ecological Restoration* 29, no. 4 (2011): 392.

⁶² Janet Marinelli, “Rain Garden Plants.” Brooklyn Botanic Garden. Accessed May 5, 2004, https://www.bbg.org/article/rain_garden_plants.

<u>Northeast Native Rain Garden Plants⁶²</u>	
<i>Athyrium filix-femina</i> , Lady fern	<i>Ilex verticillata</i> , Winterberry holly
<i>Osmunda regalis</i> , Royal fern	<i>Lindera benzoin</i> , Spicebush
<i>Osmunda cinnamomea</i> , Cinnamon fern	<i>Liquidambar styraciflua</i> , Sweet gum
<i>Carex pendula</i> , Drooping sedge	<i>Sambucus canadensis</i> , American elderberry
<i>Carex stipata</i> , Tussock sedge	<i>Viburnum dentatum</i> , Arrowwood

The daylighting of a portion of the Wingohocking Creek points to a successful example of both daylighting and partnership with the natural surrounding environments of creeks to mitigate flooding issues in Germantown. Awbury Arboretum estimates that more than 14,000,000 gallons will pass through the arboretum's watercourse and pond systems, drawing stormwater from nearby flood-prone streets to dramatically reduce the amount of pollutant overwhelming the Philadelphia's sewer system.⁶³ Although smaller areas of stream daylighting in vacant lots wouldn't operate on the same scale of water capture as the much larger area of the arboretum, by integrating bioswales and rain gardens, these lots have the potential to have a large impact despite their humble footprint.

IV. Conclusion

When addressing an issue such as deep rooted flooding in a historical neighborhood such as Germantown, we must ask ourselves what type of solution we are looking for: bandaids solutions that exist within the current system, or solutions that address the root of the problem. While flooded basements and roads might not immediately appear to be a problem with economic growth-centered capitalism, with historical context it becomes clear that the two are intrinsically linked. Should we seek to maintain the historic burial of these culverted waterways?

⁶³ "Wetlands Restoration Is an Environmental Success Story." Awbury Arboretum | Philadelphia, PA, May 20, 2021. <https://awbury.org/wetlands-article/>.

Or should we let them finally breathe, giving them the room to naturally expand and contract in daylighted areas?

This all might seem naively radical—you might gasp and whisper at the impossibility of daylighting these streams, just as the PWD representatives did. To this we say: imagining radical alternatives to status quo systems that are actively undermining the livelihoods of most residents in a neighborhood are *necessary* and *the only path forward*. If we can't imagine an alternative, an alternative will never exist. Capitalism and Western growth-centered ways of thinking are ingrained into our lives and ways of thinking, and we must directly confront and recognize these ingrained assumptions to imagine alternatives. In asking, “what does it mean for water to return?” we ask you to imagine what a livable future looks like. Whether this proposal resonates and inspires you or not, we hope that you fight for this livable future through whatever means you have.

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